

Homework 3

6.7960 Deep Learning
Fall 2024

Instructions: There are a total of 35 points for this homework. Each question is marked with its corresponding points. Some questions are **not graded**, including all bonus questions. But you are encouraged to think about and attempt them.

Submission: This assignment is **coding heavy** and will take some time. All of the code for this homework (Problems 2, 3, and 4) will be in the following colab notebook [here](#). For this assignment, do not include your code inline. Instead, export the notebook as a pdf (with the outputs) and attach the PDF to your submission. However, written answers should be included as usual. Please complete the page assignment to each section within the course site, otherwise we will apply a **3% penalty** to the homework. This can be done after you upload your submission **PDF**.

Notation: We will use this set of math notation specified on course website. For example, c is a scalar, $\mathbf{b}$ is a vector and $\mathbf{W}$ is a matrix. You are encouraged (although not enforced) to follow this notation in your typeset submission, or to the best of your ability with a handwritten response (bolding may be difficult :))

Note on use of AI Assistants: This assignment is particularly coding heavy. We understand that AI assistants can complete many coding based problems. Therefore, we would like to provide a reminder of our AI assistant policy. You may not simply ask an AI assistant to complete your code for you. This includes Google Colab autocomplete, which should not be used for completing assignments. Our full AI assistant policy can be found [here](#).

Note on Colab Autocomplete Usage: Please disable Colab's code autocomplete for this problem set. Navigate to Tools/Settings/Editor, and toggle off the option labeled "Automatically trigger code completions." For more details, refer to the instructions [here](#).

Problem 1: RNNs versus transformers (8 pt)

Recurrent neural networks, also known as RNNs, are a type of neural network used for sequence modelling. In this question, we will think conceptually about how an RNN processes information, and compare this to transformers.

Consider the simple RNN architecture shown in Figure 1. The inputs $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T$ are vectors in $\mathbb{R}^{d_{\text{in}}}$, the hidden states $\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_T$ are vectors in $\mathbb{R}^{d_{\text{hidden}}}$, and the outputs $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_T$ are vectors in $\mathbb{R}^{d_{\text{out}}}$. The hidden states and outputs are given by recurrence relations:

$$\mathbf{h}_t = \phi_h(\mathbf{W}_h \mathbf{h}_{t-1} + \mathbf{W}_x \mathbf{x}_t + \mathbf{b}_h); \quad (1)$$

$$\mathbf{y}_t = \phi_y(\mathbf{W}_y \mathbf{h}_t + \mathbf{b}_y). \quad (2)$$

The functions $\phi_h(\cdot)$ and $\phi_y(\cdot)$ are arbitrary element-wise non-linearities. The RNN has three weight matrices $\mathbf{W}_h$, $\mathbf{W}_x$ and $\mathbf{W}_y$ and two bias vectors $\mathbf{b}_h$ and $\mathbf{b}_y$.

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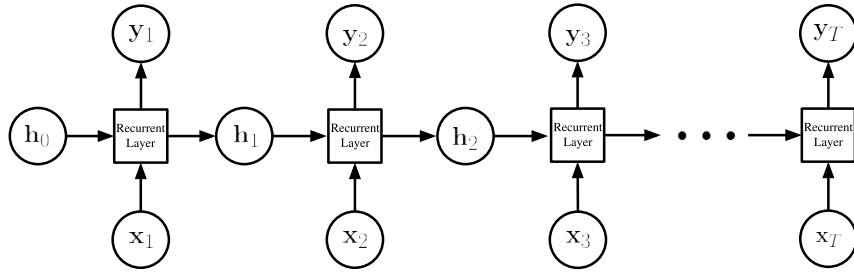


Figure 1: A simple RNN architecture. At time t , an RNN computes a hidden state $\mathbf{h}_t$ based on the current input $\mathbf{x}_t$ and prior hidden state $\mathbf{h}_{t-1}$. The RNN also spits out an output $\mathbf{y}_t$.

For simplicity, in this question we will set the initial hidden state $\mathbf{h}_0 = \mathbf{0}$, we will set the non-linearity ϕ_h to the identity $\phi_h(\mathbf{h}) = \mathbf{h}$ and we will set the biases $\mathbf{b}_h = \mathbf{0}$ and $\mathbf{b}_y = \mathbf{0}$. Under this simplification, after one time step $T=1$: $\mathbf{h}_1 = \mathbf{W}_x \mathbf{x}_1$ and $\mathbf{y}_1 = \phi_y(\mathbf{W}_y \mathbf{W}_x \mathbf{x}_1)$.

- (1pt)** Derive formulae for hidden state $\mathbf{h}_2$ and output $\mathbf{y}_2$ in terms of the weight matrices $\mathbf{W}_y$, $\mathbf{W}_x$, $\mathbf{W}_h$ and inputs $\mathbf{x}_1, \mathbf{x}_2$.
- (1pt)** Derive formulae for hidden state $\mathbf{h}_3$ and output $\mathbf{y}_3$ in terms of the weight matrices $\mathbf{W}_y$, $\mathbf{W}_x$, $\mathbf{W}_h$ and inputs $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$.
- (1pt)** Derive formulae for hidden state $\mathbf{h}_T$ and output $\mathbf{y}_T$ in terms of the weight matrices $\mathbf{W}_y$, $\mathbf{W}_x$, $\mathbf{W}_h$ and inputs $\mathbf{x}_1, \dots, \mathbf{x}_T$.
- (1pt)** Suppose the sequence length T is very long. What do you notice about the contribution of the first input $\mathbf{x}_1$ to the last output $\mathbf{y}_T$ of the RNN?

Another way of handling sequential data is to use a self-attention layer, à la transformers. Given a sequence of inputs $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T$. A self-attention layer computes:

$$\text{pair-wise inner products:} \quad \alpha_{ij} = \frac{1}{\sqrt{d}} (\mathbf{Q} \mathbf{x}_i)^\top (\mathbf{K} \mathbf{x}_j) \quad \text{for } i = 1, \dots, T \text{ and } j = 1, \dots, T; \quad (3)$$

$$\text{outputs:} \quad \mathbf{y}_t = \sum_{j=1}^T \frac{e^{\alpha_{tj}} \mathbf{V} \mathbf{x}_j}{\sum_{j=1}^T e^{\alpha_{tj}}} \quad \text{for } t = 1, \dots, T, \quad (4)$$

where $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ are the *query*, *key* and *value* matrices and d is the embedding dimension.

- (3pt)** For the first two questions, your answer only needs to indicate the asymptotic scaling with sequence length T . Use big- $\mathcal{O}$ notation, and ignore any other factors.

- For RNNs, how many floating point operations are needed for a forward pass?

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- For a self-attention layer, how many floating point operations are needed for a forward pass?
- With sufficient parallel hardware, will performing a forward pass on a transformer or RNN be faster? Why is this the case?

Hint: Think about different ways to arrange the computation of Equations 3 and 4.

- (f) **(1pt)** If you wanted to train and deploy a neural network that operates on very long sequences $T \rightarrow \infty$, would you rather use an RNN or a transformer? Why?

Hint: There are different possible answers here, and we are just looking for some short sensible commentary that reflects on memory and time complexity mentioned in previous problems.

Problem 2: Implementing a Transformer (11 pt)

In this problem, you'll implement a Transformer (from scratch!). We'll first focus on writing a self-attention module. In the next two problems we'll explore how to use transformers for two entirely different domains: image classification and language. Throughout this problem, let B be the batch size and T be the sequence length.

These problems will be implemented using the following colab notebook [here](#). These problems are coding heavy and will take some amount of time. For the submission, **download the entire colab notebook as a PDF document** and attach this PDF to your solution at the end. No need to copy and paste the code in-line. However, if we ask for a non-coding response, answer the question here.

- (a) **(4pt)** Our first task is to implement single-headed self-attention. Take a look at the class `AttentionHead`. For an input $x \in \mathbb{R}^{T \times d}$, we want to setup three linear layers $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d \times d_k}$. Use these to create your queries, keys, and values.

$$\mathbf{Q} = \mathbf{x}\mathbf{W}_Q \quad \mathbf{K} = \mathbf{x}\mathbf{W}_K \quad \mathbf{V} = \mathbf{x}\mathbf{W}_v$$

Then compute your attention weights

$$\mathbf{A} = \text{softmax}(\mathbf{Q}\mathbf{K}^T / \sqrt{d_k})$$

Here $\mathbf{A}_{ij}$ indicates the weight of token j when computing the new representation for token i . Finally use your attention weights to compute your output as a weighted combination of the values:

$$\text{Out} = \mathbf{A}\mathbf{V}$$

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Deliverable Implement AttentionHead. Notice that the forward function takes an argument called `attn_mask` $\in \{0, 1\}^{T \times T}$. If `attn_maski,j` = 0 token i should not attend on token j (e.g., `Aij` = 0). Modify your code accordingly.

Hint: If you want token i not to attend on token j what value should you set in the input of the softmax?)

- (b) **(3pt)** We now implement multi-headed attention.

Take a look at the class `MultiHeadedAttention`. Our output should separately compute the attention outputs for each of the heads, then concatenate the outputs together and project them into the output dimension. Specifically for parameter matrix W_O (what size should W_O be?)

$$\text{MultiHead}(x) = \text{Concat}(\text{head}_1(x), \dots, \text{head}_n(x))W_O$$

Deliverable Implement `MultiHeadedAttention`.¹

- (c) **(3pt)** Now let's put it all together!

Take a look at the feed-forward-network module `FFN` and the residual model `AttentionResidual`. We've already implemented this bit for you (you're welcome!). The `FFN` performs normalization along with linear layers interspersed by GELUs² (similar to the MLPs you've seen in previous PSETs). `AttentionResidual` then sends the input through both multiheaded attention and the `FFN`, adding a residual after each step.

Deliverable Implement `Transformer`, which passes the input through successive `AttentionResidual` layers.

Make sure you can pass the given test cases (they just check output sizes). In particular, test case 3 checks whether you've dealt with `attn_mask` correctly. The provided attention mask indicates that each token should only attend to itself and the token before it. Your returned `A` should reflect this attention pattern.

- (d) ~~**(1pt)**~~ Transformers are typically designed to handle discrete token sequences, like words in a sentence. However, in many domains like audio and images, the input is continuous rather than discrete.

Deliverable Please restrict your answer to 2-3 sentences in 1 paragraph. Explain how the transformer architecture can be adapted to handle continuous inputs, such as audio and images. Think about how tokenization, embedding, and positional

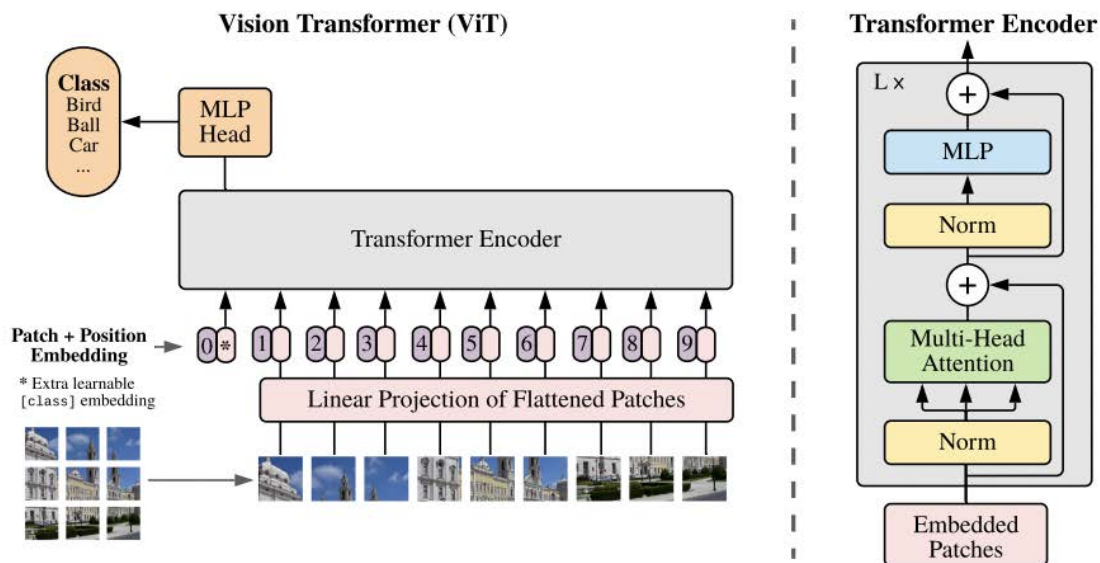
¹Note, in most industrial settings, rather than looping over the heads sequentially, all the heads are computed at the same time in a vectorized fashion. This provides a speedup by letting you compute the heads in parallel, but don't worry about this here.

²a Gaussian Error Linear Unit (GELU) functions here as a smoother version of a ReLU.

encodings need to be adjusted to effectively apply self-attention to these types of inputs. Consider how you would divide continuous data into meaningful "tokens" and how the transformer can capture local and global relationships within this data.

Problem 3: Vision Transformers (6 pt)

So far, the best performing model we've seen for image classification has been CNNs. Recently, transformer architectures have been shown to have similar to better performance. One such model called Vision Transformer (ViT) splits up images into regularly sized patches (Dosovitskiy et al. [2020]). The patches are treated as a sequence and attention weights are learned as in a standard transformer model.



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Figure 2: ViT Architecture, Figure from Dosovitskiy et al. [2020]

In detail the ViT has a few steps (see Figure 2).

- First we embed the patches using into a sequences of embeddings.
- We add a positional encoding to the embedding which captures the position of each patch in the image.
- We prepend an extra learned class embedding to our sequence and pass the entire sequence through a transformer.

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- We extract the final representation of the class embedding and learn a linear layer (MLP Head) to predict the probability of each class.
- We supervise the class with cross entropy loss.

Now that we've implemented a transformer, we can use it to implement a ViT! Make sure you've already done the previous section.

- (a) **(2pt)** We first implement our patch embedding. Take a look at the class `PatchEmbed`. For a given image, we want to split the image into square patches. **Each patch should then be flattened and linearly projected with some weight.**

For example, suppose we want embeddings of size 128. If your image is size $(3, 32, 32)$ and your patches are 4×4 , you should end up with 64 patches. Flattened, each patch contains $3 * 4 * 4 = 48$ elements. **We want to learn a linear projection from those 48 elements to our output dimension 128.** We'll then end up with a sequence of 64 inputs of 128 elements each to pass into our transformer!

Deliverable Implement `PatchEmbed`.

Hint: Splitting up the patches manually and then using `nn.Linear` will be painful. Instead, look at `nn.Conv2d`. How can you use this to implement the patch embedding?

- (b) **(1pt)** Read through the `VisionTransformer` (implemented for you). Take a look at the *positional embedding*. Positional embeddings encode the position of each element in the sequence. In this case, the positional embeddings for every position in the sequence is *learned*. However, this creates a strict limit on how many tokens can be passed to the transformer (if you only had 64 position embeddings, the positional embedding of the 65th token is undefined!)

Suppose you wanted to implement a transformer that can take arbitrarily long inputs (ignore any memory or time constraints). Describe a way to implement the positional embedding such that there is no maximum sequence size.

- (c) **(1pt)** Train the Vision Transformer on CIFAR-10! We've implemented the training loop for you. Run the cells to train a model and **report your validation accuracy here** (it should be greater than 50%). This should take about 5 minutes.
- (d) **(1pt)** The attention maps for transformers tell us which patch relied on which other patch. Let's take a look at the attention heatmap of the class token (averaged over all heads and layers). At a high level this can give us a sense of which parts of the image the model is relying on. We provide code to visualize this heatmap for 10 validation images.

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Include the images with their heatmaps here. What do you observe about the heatmaps?

- (e) **(1pt)** Previously, we have seen convolutional neural networks (CNNs) used for image classification. Answer the following questions with regards to the capabilities of CNNs and ViTs. We expect no more than a few sentences for each question.
- CNNs have inductive biases that emphasize local spatial patterns by design, while ViTs rely on global attention with limited biases. How does this difference affect their abilities, particularly on small vs. large datasets?
 - CNNs capture spatial information due to their network structure, while ViTs generally rely on learned positional encodings. How do these approaches impact each model's ability to generalize to images of different sizes?

Problem 4: DialogueGPT (10 pt)

Now let's use our Transformer to train a language model! We're going to train a language model on some Shakespeare. Run the cell to download `input.txt` which contains some Shakespeare text. Each element in `all_dialogues` will be one example in our dataset.

- (a) **(2pt)** Our first step is to build a *tokenizer*, which splits line of dialogue into individual tokens and assigns each token to an ID. We're going to use NLTK's `word_tokenize`, which splits words and punctuation into their own tokens.

Take a look at `MyTokenizer`. This tokenizer has three special tokens: `start` (which will start every example), `pad` (used to pad examples to the same length), and `unk` (used when encountering a word not in our vocabulary). We've initialized the tokenizer for you.

Deliverable Implement the following functions in `MyTokenizer`:

- `encode`: convert a string to a series of token ids and prepend the token id for the start token. Use `word_tokenize` to split the string.
- `decode`: convert an array of token ids back into tokens. Join them with a space.

Make sure that your tokenizer fulfills the test case.

- (b) Read over and make sure you understand how we create the `DialogueDataset` and data loader. The data loader pads the list of tokens such that they are all the same length. It outputs a dictionary with two elements:
- `input_ids` contains the input ids for each element in the batch (padded to the right to the max length)

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- `input_mask` indicates which tokens are pad tokens (and should thus be ignored).

Deliverable You don't need to do anything for this question.

- (c) **(2pt)** Time to implement DialogueGPT! Review the lecture on language models if you haven't already.

Deliverable Fill out TODOs in the `__init__` and forward methods. The forward call should:

- Given the token ids, retrieve the corresponding token embeddings. Add to this a **learned positional embedding**.
 - Generate a *causal attention mask*. Remember that for GPT, every token only depends on itself and the tokens before it
 - Pass the embeddings and the attention mask to the transformer and the language model head. Output logits of size $(T \times V)$ where T is the number of tokens and V is the vocabulary size. This step is implemented for you
- (d) **(2pt)** Let's implement the loss. Remember that a language model is trained to predict the next token given a prefix of tokens.

Suppose our vocab size is V and we have T tokens in our training example (including the start token). Then our model will output logits $O \in \mathbb{R}^{T \times V}$. Our loss for this example will be $T - 1$ individual classification losses, where using logit vector $O[i]$ we will predict the token id for token $i + 1$ via **cross entropy loss**. This means we will not supervise the start token, nor will we use the last logit vector $O[-1]$. An illustration of this is shown below.



Figure 3: Illustration of GPT Loss

Deliverable Implement DialogueLoss. Remember to take into account the `inp_mask` to ignore supervising tokens that correspond to padding.

- (e) **(1pt)** Go back to DialogueGPT read the `generate` function, which we have implemented for you. The `generate` function takes in a prefix of token ids and autoregressively generates `num_tokens` more tokens, by greedily picking the most likely next token and adding it back to the input.

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Generating T tokens is often much slower than training on an input with T tokens. Comment on why this is the case.

- (f) **(1pt)** Now its time to train DialogueGPT! Run the cells to train the model. This step will take you around 30 minutes, so budget accordingly. Note: if you're having trouble debugging, try overfitting to just a few examples (e.g., make your training dataset size 10 or so).

The training code generates some text after every epoch. What do you notice about the generations as the epochs progress?

- (g) **(1pt)** Generate 50 tokens of input text. Our language model is pretty small and hasn't been trained for very long, but you should still get something approximating english. Paste the output of your model here.
- (h) **(0.5pt)** As you can see above, GPT-like models often struggle with generating long, coherent outputs and tend to produce repetitive or degenerate sequences when generating many tokens. One common solution to this issue is to use decoding strategies such as nucleus sampling instead of greedy decoding.

Deliverable Please restrict your answer to 2-3 sentences in 1 paragraph. Explain how nucleus sampling works and why it may produce more diverse and coherent outputs compared to greedy decoding. What trade-offs does it introduce in terms of generation speed and output quality?

Hint: Nucleus sampling, also known as top- p sampling, involves selecting from a subset of tokens whose cumulative probability exceeds a threshold (often denoted as p). Instead of always selecting the token with the highest probability (as in greedy decoding), the model samples from a dynamically sized pool of likely tokens [Holtzman et al. \[2020\]](#).

- (i) **(0.5pt)** As you can see from the forward pass of DialogueGPT, generating a sequence of tokens requires multiple forward passes through the model, where each new token depends on all previously generated tokens. This can become computationally expensive, especially for long sequences, because the model needs to compute attention over the entire sequence at each step.

One common solution to make this process more efficient is key-value (KV) caching, which allows us to store and reuse previously computed values, reducing the need to recompute them for every new token.

Deliverable Please restrict your answer to 2-3 sentences in 1 paragraph. On a high-level, explain in simple terms why KV caching is useful in DialogueGPT when generating long sequences. How does it help reduce the amount of work the model

needs to do at each step? Why is this important for making the generation faster, especially for longer sequences?

Hint: Without KV caching, the model would have to recompute the attention for all tokens, not just the new one. KV caching avoids this extra computation. Reusing key-value pairs improves efficiency by storing past hidden states and retrieving them when needed, which greatly accelerates the generation process, particularly for long conversations or sequences. This method is especially critical in transformer models, where self-attention requires computing interactions between all tokens [Brown \[2020\]](#).

References

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